GRISHMA BALGI

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Modelling, Simulation & Controls

EDUCATION

New York University | May 2027
M.S. Mechatronics and Robotics | GPA: 3.834/4.0

Veermata Jijabai Technological Institute | June 2025
B.Tech. Mechanical Engineering; Honors in Automobile Engineering | GPA: 9.04/10

TECHNICAL SKILLS

Programming: Python, MATLAB, NumPy

Modelling: MuJoCo, Altair romAI/Twin Activate, Creo, Ansys

Methods: Rigid-body modelling, kinematics, passive dynamics, parameter tuning, sensor-data analysis

EXPERIENCE

Prema Lab, New York University | Sep. 2025 - Present
Graduate Research Assistant | Robotics fabrication, instrumentation and simulation

- Developing an 18-segment model of a fabricated TPU 95A tentacle in MuJoCo, representing passive bending with effective joint stiffness, damping and tendon elements.
- Iterating on geometry and passive response; investigating fluid-force effects while separating provisional model parameters from quantities requiring physical calibration.
- Fabricated a robotic-fin test rig and automated Silhouette Portrait 4 cutting through Python USB commands, generating coordinate-based paths for candidate fin geometries.
- Developed a motor-control interface and Arduino acquisition firmware to vary oscillation waveform, amplitude and frequency; logged load-cell readings alongside commanded and measured motor positions.

Schneider Electric | June 2024 - June 2025
Engineering Intern | Mechanical modelling and sensor systems

- Created mechanical models and explored rigid-body dynamics and simulation-derived reduced-order models using Creo, Ansys and Altair romAI/Twin Activate.
- Used simulation data to model relationships between strain, stress and load; investigated reduced-order modelling for circuit-breaker mechanisms.

SELECTED PROJECTS

UR10e Kinematics in MuJoCo
Python, NumPy, MuJoCo, numerical methods | Robotics coursework

- Implemented forward-position kinematics and numerical inverse-position solutions for a UR10e model within a supplied MuJoCo coursework framework.
- Explored velocity kinematics and pseudoinverse-based joint-velocity calculations, visualising robot motion in simulation.